Matrices (AI HL)

ANSWER KEY

Section 1: Matrix Multiplication and Inverses

1 Compute the matrix product AB for $A = [[1, 2], [3, 4]]$ and $B = [[2, 0], [1, 3]]$.

a. $[[4, 6], [10, 12]]$ ✓

b. $[[2, 0], [3, 12]]$

c. $[[4, 10], [6, 12]]$

d. $[[3, 2], [4, 7]]$

Row 1: $[1(2) + 2(1), 1(0) + 2(3)] = [4, 6]$. Row 2: $[3(2) + 4(1), 3(0) + 4(3)] = [10, 12]$.

2 Find the inverse of the matrix $M = [[3, 5], [1, 2]]$.

a. $[[2, -5], [-1, 3]]$ ✓

b. $[[-2, 5], [1, -3]]$

c. $[[3, -1], [-5, 2]]$

d. $[[1/3, 1/5], [1, 1/2]]$

$\det(M) = 3(2) - 5(1) = 6 - 5 = 1$. The inverse is $\frac{1}{\det}[[d, -b], [-c, a]] = [[2, -5], [-1, 3]]$.

3 If A and B are invertible matrices of the same dimension, what is $(AB)^{-1}$?

a. $B^{-1}A^{-1}$ ✓

b. $A^{-1}B^{-1}$

c. BA^{-1}

d. AB^{-1}

The socks-and-shoes property of matrix inverses reverses order: $(AB)^{-1} = B^{-1}A^{-1}$.

Section 2: Determinants and Singularity

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4 For what value of k is the matrix $A = [[k, 6], [2, 3]]$ singular (non-invertible)?

a. $k = 4$ ✓

b. $k = 6$

c. $k = 0$

d. $k = 9$

A matrix is singular when its determinant is zero: $\det(A) = 3k - 12 = 0 \Rightarrow 3k = 12 \Rightarrow k = 4$.

5 If A is a 3×3 matrix with $\det(A) = 5$, what is $\det(2A)$?

a. 40 ✓

b. 10

c. 20

d. 80

For an $n \times n$ matrix, $\det(cA) = c^n \det(A)$. Here $n = 3$, so $\det(2A) = 2^3 \times 5 = 8 \times 5 = 40$.

6 If $\det(A) = 4$, what is $\det(A^{-1})$?

a. $\frac{1}{4}$ ✓

b. -4

c. 4

d. 16

$\det(A^{-1}) = \frac{1}{\det(A)} = \frac{1}{4}$.

Section 3: Eigenvalues and Eigenvectors

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7 Find the eigenvalues of the matrix $A = [[4, 2], [1, 3]]$.

a. $\lambda = 5$ **and** $\lambda = 2$ ✓

b. $\lambda = 4$ and $\lambda = 3$

c. $\lambda = 6$ and $\lambda = 1$

d. $\lambda = 7$ and $\lambda = 0$

Characteristic equation:
$\det(A - \lambda I) = (4 - \lambda)(3 - \lambda) - 2 = \lambda^2 - 7\lambda + 12 - 2 = \lambda^2 - 7\lambda + 10 = 0 \Rightarrow (\lambda - 5)(\lambda - 2) = 0$. Roots are $\lambda = 5, 2$.

8 Find an eigenvector corresponding to $\lambda = 5$ for $A = [[4, 2], [1, 3]]$.

a. $[2, 1]^T$ ✓

b. $[1, 2]^T$

c. $[1, -1]^T$

d. $[2, -1]^T$

Solve $(A - 5I)\mathbf{v} = \mathbf{0} \Rightarrow [[-1, 2], [1, -2]][x, y]^T = [0, 0]^T \Rightarrow -x + 2y = 0 \Rightarrow x = 2y$. With $y = 1$, $\mathbf{v} = [2, 1]^T$.

9 What is the sum of the eigenvalues of an $n \times n$ matrix A equal to?

a. **The trace of A (sum of diagonal entries)** ✓

b. The determinant of A

c. The rank of A

d. n

By algebraic properties of the characteristic polynomial, $\sum \lambda_i = \text{tr}(A)$.

Section 4: Diagonalization & Matrix Powers

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10 If a matrix A is diagonalized as $A = PDP^{-1}$, how is the power A^n computed efficiently?

a. $A^n = PD^nP^{-1}$ ✓

b. $A^n = P^nD^n(P^{-1})^n$

c. $A^n = D^n$

d. $A^n = PDP^{-1}n$

$A^n = (PDP^{-1})(PDP^{-1})\ldots = PD^nP^{-1}$. For a diagonal matrix D, D^n simply raises each diagonal entry to the nth power.

11 If $D = [[2, 0], [0, 3]]$, what is D^4?

a. $[[16, 0], [0, 81]]$ ✓

b. $[[8, 0], [0, 12]]$

c. $[[24, 0], [0, 24]]$

d. $[[16, 81], [0, 0]]$

For diagonal matrices, powers are computed element-wise along the diagonal: $2^4 = 16$ and $3^4 = 81$, yielding $[[16, 0], [0, 81]]$.

12 When is an $n \times n$ matrix A guaranteed to be diagonalizable?

a. **When it has n linearly independent eigenvectors (or n distinct eigenvalues)** ✓

b. When its determinant is non-zero

c. Only if it is a symmetric identity matrix

d. When all its entries are positive

A matrix is diagonalizable if and only if there exists an eigenbasis consisting of n linearly independent eigenvectors.

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